Artificial intelligence (AI) systems are increasingly embedded in the tools people use to make decisions, perform tasks, and access information. Yet, whether people understand their value and whether they recognize their limitations remains an important challenge.

I am a Human-Computer Interaction (HCI) researcher studying human-centered AI, with a focus on how **interaction techniques shape people's perceptions**, mental models, and reliance on AI systems. Drawing on cognitive science, decision-making theories, and empirical HCI methods, I investigate how people interpret AI-generated information and incorporate it into their decisions. My research combines the development of **computational techniques and interactive prototypes** with empirical evaluations with end users. Through this work, I aim to design AI systems that enable people to **move beyond passively consuming AI outputs** and participate meaningfully in human-AI collaboration.

My research has developed along three interconnected directions: 1) developing interactive AI systems that support human understanding and control, 2) designing computational and interaction techniques for appropriate reliance on AI, and 3) investigating human-AI collaboration in emerging generative AI applications. Across these directions, **I employ mixed-methods approaches**, including qualitative interviews, controlled experiments, surveys, and analysis of behavioral interaction data. I also develop **computational techniques and interactive prototypes** to investigate how people engage with AI systems in real-world contexts. My work has appeared in leading HCI venues, including CHI (**Honorable Mention Award**), IJHCS(**Top-Cited for the Journal**), UMAP, ASSETS, and HCII.

Understanding Human Mental Models and Perceptions of AI

Understanding how people interpret and make sense of AI systems is fundamental to explaining their decisions and behavior. People do not interact with AI solely based on its technical capabilities; their decisions are shaped by their expectations, prior knowledge, and understanding of what AI can and cannot do. My research investigates how integrating AI into existing work practices shapes **people's perceptions, mental models, and willingness** to incorporate AI-generated information into their decisions. Further, I investigate how interactive computational systems can allow integrating human knowledge into machine learning workflows.

Through in-depth **qualitative and phenomenological inquiries**, I examined the challenges business decision-makers encounter when applying AI in professional settings, including the underlying reasons for resistance to AI adoption [1]. Extending these inquiries, I designed interviews, questionnaires, and interactive research prototypes **informed by sensemaking perspectives** to investigate how people interpret AI-enabled analytics [2]. In this context, I developed interactive prototypes that allowed users to evaluate AI models for customer classification and explore how model outputs related to their existing knowledge and practices [3].

Through a **research collaboration with Sappi Europe**, the extensions of prototypes I developed were embedded within real-world customer analytics workflows. Evaluations revealed that non-technical users, including sales professionals, often struggled with limited control over AI models and wanted more intuitive ways to incorporate their domain knowledge into analytical processes. These findings highlight an important cognitive dimension of human-AI interaction: people need opportunities to connect AI-generated information with their existing knowledge and mental models to make sense of outputs and determine whether they are appropriate for a given decision. Subsequent improvements in these tools supported users in employing a transparent approach to applying AI accountably. Altogether, these studies demonstrated that interactions that keep people in control of AI use within their existing work enable them to integrate AI consciously to improve their decisions.

Designing Human-AI Interaction for People's Reliance on AI

As AI systems become increasingly integrated into decision-making, users must determine when to accept or reject AI-generated recommendations. However, when AI systems suffer from random or systematic errors, **inappropriate reliance can lead to poor decisions**. *Appropriate reliance* can be defined as people accepting its assistance only when it is correct and not otherwise. My research examines the interactional and computational factors that influence people's reliance on AI and investigates how system design can support more informed human-AI decision-making. In this context, I analyzed these factors to understand

and isolate their influence on people's appropriate reliance on AI. This outcome of this work was published at CHI, receiving an **honorable mention award** [4].

Recognizing that identifying the factors influencing reliance is not sufficient, I subsequently investigated computational and interaction techniques to encourage appropriate AI use. Through controlled empirical studies, I examined interventions involving reflective nudges, model performance indicators, uncertainty information, and users' self-assessed confidence in their decisions. Analysis of interaction data showed that these interventions encouraged users to engage more deeply with AI systems and explore uncertainty and explanation features [5]. However, reflections did not significantly impact people's decisions when AI models underperformed or had uncalibrated confidence.

Extending this work, I designed exploratory interactions that allowed people to investigate how AI models worked, modify inputs and datasets, and identify model blind spots. The evaluation showed that active engagement with AI systems encouraged more cautious behavior and better decision-making [6]. Engaging users can be a critical factor in improving their reliance on AI decision support. This line of work also examined the value of understanding models, supporting self-correction, and analytically engaging with decision support.

Examining the Value of Human-AI Interaction

AI systems can position people as passive *consumers* or enable them to *actively* participate in shaping system outputs. I argue that meaningful human-AI collaboration requires interactions that are *explicit, informed, and intentional*, allowing people to initiate interactions, retain control, and remain engaged throughout the process. To establish this perspective, I developed a systematic framework distinguishing active human-AI interaction from passive consumption of AI outputs. My analysis of human-centered AI research showed that most AI systems limit user interactions beyond passive consumption. The outcome of this work was published in IJHCS and is currently the **highest-cited paper for the journal** [7], providing a conceptual foundation for designing systems that preserve human agency.

The recent progress in generative AI systems has improved user interaction, co-creation, and task execution with AI. Evaluating how AI support improves people's work, for instance, in creating outputs with generative methods, can inform what value it brings. Aligned with this rationale, I examined the value of generative AI in writing alternative text for science figures. Applying mixed methods (e.g., tool usage and interviews), our evaluation showed that people place high trust in and rely on AI systems without oversight, leading to concerns about improper integration of AI in complex workflows. Therefore, **designing systems that integrate oversight** within themselves is highly needed. This work is partly published at ASSETS [8] and is currently under review at TACCESS [9].

Extending this line of work, my current work examines **how persuasive design influences users' reliance** on AI, which can impact whether they use AI more analytically or apply oversight. I am also investigating interactive analytics with generative AI, focusing on how people can evaluate and modify AI outputs. Understanding how people engage with AI and data analytics, I aim to design interactions that support their analytical evaluation of outputs. Initial evidence from my current work shows that interaction facilitated by evaluations of outputs can shape the cognitive processes through which people interpret information, evaluate alternatives, and make decisions.

Research Agenda

My research goal is to advance the understanding of human cognition and apply this understanding to the design of AI experiences and decision-support systems. Below, I briefly outline this research agenda.

Designing Interactive AI Experiences. As generative AI becomes increasingly capable of responding to natural-language requests and carrying out multi-step tasks, it is important to understand how people communicate their intentions, evaluate system responses, and retain control over AI-assisted processes. Building on my experience developing interactive machine-learning tools, I plan to design and **prototype interactive and conversational AI systems** that support people in exploring information, evaluating generated outputs, and incorporating their own knowledge into AI-assisted tasks. I am also interested in extending use cases beyond work settings to personal contexts (e.g., emotional and psychological) and to diverse user groups (e.g., laypeople, students, and people with different skills/abilities).

Designing to Curb Misuse of AI. Despite growing AI adoption, disparities in people's ability to influence and be influenced by AI systems are widening. Misuse of AI systems can have significant consequences in diverse domains (e.g., healthcare, business intelligence), ranging from basic information extraction tasks to critical decision-making. Applying mixed-methods studies, my future work will develop frameworks and decision-support tools that help people use AI responsibly and appropriately. I will examine how different forms of interaction influence cognitive engagement, confidence calibration, metacognitive awareness, and decision quality. The findings will inform practical guidelines for designing AI systems and developing institutional practices that support appropriate AI use. My investigation could also include how people's misuse of AI is influenced by public discourse or online messaging on AI prowess.

Extending Perspectives. AI systems are increasingly used by people with different levels of expertise, resources, and access to technology. Their needs and experiences are shaped by the social, cultural, and institutional contexts in which these systems are introduced. I am interested in context-sensitive research conducted in collaboration with communities and institutions. Such work can reveal how local practices, user expectations, and institutional constraints shape human-AI interaction, while informing systems that are responsive to the people and communities they serve.

Across these directions, I bring experience implementing **interactive (generative) AI prototypes**, designing and conducting empirical studies with people to evaluate their use. Having spent most of my life in an **under-resourced region of the Indian subcontinent**, I bring a personal understanding of how differences in access to technology, infrastructure, and institutional resources shape people's experiences with digital systems. I have **mentored several undergraduate and graduate students**, including in academic and professional settings.

References

[1] Muhammad Raees, Ioanna Lykourentzou, Vassilis-Javed Khan, and Konstantinos Papangelis. (2026). "We Still Use Spreadsheets." Understanding Business Decision-Makers' Perceptions and Barriers to AI Analytics. *CHI EA 2026*. doi: 10.1145/3772363.3798820

[2] Muhammad Raees, Vassilis-Javed Khan, and Konstantinos Papangelis. (2023). Four Challenges for IML Designers: Lessons of an Interactive Customer Segmentation Prototype in a Global Manufacturing Company. *CHI EA 2023*. doi: 10.1145/3544549.3585788

[3] Muhammad Raees, Vassilis-Javed Khan, and Konstantinos Papangelis. (2025). UX Challenges in Implementing an Interactive B2B Customer Segmentation Tool. *CHI EA 2025* doi: 10.1145/3706599.3706687

[4] Muhammad Raees, Vassilis-Javed Khan, Ioanna Lykourentzou, and Konstantinos Papangelis. (2026). Do People Appropriately Rely on AI-Advice? An Analytical Review of HCI Research on Human-AI Decision-Making. *CHI 2026*. Honorable Mention Award. doi: 10.1145/3772318.3791467

[5] Muhammad Raees, Vassilis-Javed Khan and Konstantinos Papangelis. (2025). Exploring Persuasive Engagement to Reduce Over-Reliance on AI-Assistance in a Customer Classification Case. *UMAP 2025*. doi: 10.1145/3699682.3728320

[6] Muhammad Raees and Konstantinos Papangelis. (2026). Understanding Interactive Model Engagement for Appropriate Reliance: An Exploration of AI-Assisted Decision Support in Customer Classification. *CHI EA 2026*. doi: 10.1145/3772363.3798834

[7] Muhammad Raees, Inge Meijerink, Ioanna Lykourentzou, Vassilis-Javed Khan, and Konstantinos Papangelis. (2024). From explainable to interactive AI: A literature review on current trends in human-AI interaction. *IJHCS*. 189, 103301. Most cited on IJHCS. doi: 10.1016/j.ijhcs.2024.103301

[8] Muhammad Raees*, Yugo Iwamoto*, Jamison Heard and Garreth W. Tigwell. (2025). Exploring Generative AI to Support Disability Service Professionals in Writing Image Descriptions for HCI Science Figures. *ASSETS 2025*. doi: 10.1145/3663547.3759763

[9] Muhammad Raees, Yugo Iwamoto, Konstantions Papangelis, Jamison Heard and Garreth W. Tigwell. (2026). How University Disability Services Professionals Write Image Descriptions for HCI Figures Using Generative AI. *Under review at TACCESS*. doi: 10.48550/arXiv.2602.08937